

Handling Occlusions via Occlusion-aware Labels

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Abstract

In real-world scenarios, many objects in view are often partially occluded, making the ability to handle occlusion essential for everyday activities. While human vision exhibits robustness to extreme occlusion, Convolutional Neural Networks struggle in this regard. Current regional masking strategies effectively improve generalization to occlusion. However, these methods typically eliminate informative pixels in training images by overlaying a patch of either random pixels or Gaussian noise, leading to a loss of image features. This limitation can be addressed by employing a more informative label. Rather than augmenting occluded images during training and assigning identical labels to both clean and occluded images, we differentiate between their labels. Essentially, we treat occlusion as a virtual class and assign it a virtual label. During training with occluded images, we merge the ground truth labels with these virtual labels, thereby informing the model about the presence of occlusion. Our findings indicate a 49.26% improvement in generalization for occlusion scenarios and an 8.22% enhancement on the common corruptions benchmark.

Keywords: Deep Neural Networks, Augmentations, Generalization, Occlusion

1 Introduction

The capability of a system, be it a neural network or a biological brain, to effectively generalize to unseen scenarios depends on its inherent inductive bias (Mitchell, 1980; Goyal and Bengio, 2022). Once established, this bias guides the selection of a specific decision boundary or hypothesis from the model's hypothesis space based on the training dataset (Meding et al., 2021).

Convolutional neural networks (CNNs) are designed with an inductive bias that assumes spatial hierarchies in data (LeCun et al., 2015). CNNs thus automatically learn features from training data, eliminating the need for hand-crafted features. While shape and texture are two complementary cues for recognizing objects, it has been shown that the features learned by CNNs often exhibit a preference for either shape or texture, mainly due to the statistics of the training data (Geirhos et al., 2018; Li et al., 2020). Employing data augmentation—by creating modified versions of the existing data—can affect the characteristics of the dataset, helping to reduce bias and improve the model's ability to generalize to out-of-distribution images (Hermann, 2022).

Within a supervised setting, the labels associated with training data can also serve as an inductive bias (Beyer et al. (2020); Tsipras et al. (2020)). For instance, depending on the assigned label, models may exhibit biases towards shape (Geirhos et al. (2018)), texture, or both shape and texture concurrently (Li et al. (2020)), which could result in biased models. In our experimentation with augmentation techniques like noise, we have observed that this also extends to label assignment for transformed data during training. Instead of adhering strictly to conventional augmentation methods of simply augmenting a set of transformed data in the training pipeline, we find it more effective to adjust the labels. This is what we refer to as Virtual Fusion (VF). Essentially, VF-training treats each of the augmentations employed during training as a virtual class, and when any transformed data is used during training, the labels are fused simultaneously. As individuals, we are accustomed to recognizing changes in images and distinguishing between an image and its noisy rendition. VF aims to achieve this alignment through labels; this is more effective in enhancing model robustness and improving generalization compared to standard data augmentation. As a point of reference, Table 1 presents a comparison



Figure 1: Methods for occlusion: Traditional augmentations maintain label consistency despite noise. Cutmix uses convex combinations of labels, but no noise is used during training. VF treats noises as virtual classes, adjusting labels when occluded images are used in training.

of normal augmentation and VF-training with three types of image transformations during training. Normal augmentation enhances baseline corruption errors by 14.31%, whereas VF-training improves them by 34.01%.

In this study, we evaluate the efficacy of VF in scenarios involving occlusions. Occlusion differs from typical image distortions found in common corruption benchmarks (Hendrycks and Dietterich, 2019) such as brightness, Gaussian noise, and zoom blur, as it results in the loss of features/information within the queried image. The human visual system robustly distinguishes between whole and partially occluded views. When comparing images and their occluded counterparts, we recognize that something is covering the view. With VF, we can convey similar information to the model during training with occluded images. This proves to be more effective in achieving lower error rates under occlusion compared to standard masked image augmentations.

2 Related Works

CNNs demonstrate a decreased robustness to partial occlusion in contrast to human perception (Zhu et al., 2019). Previous research has addressed the challenge of occlusion across various tasks such as Object classification Kortylewski et al. (2020); Zhang et al. (2023a), tracking Van Hoorick et al. (2023); Wang et al. (2024), instance segmentation Ke et al. (2021); Qiu et al. (2024), and pose estimation Zhang et al. (2023b).

The existing augmentation techniques associated with occlusion include Cutout (DeVries and Taylor, 2017), Cutmix (Yun et al., 2019), Random Erasing (Zhong et al., 2020), and Patch Gaussian (Lopes et al., 2019). Cutout involves removing sections of input images and augmenting data with partially occluded versions of existing samples (DeVries and Taylor, 2017). Similarly, Random Erasing, randomly selects a rectangular region in an image and replaces its pixels with random values within the range of $[0, 255]$ (Zhong et al., 2020). Patch Gaussian follows a similar approach but it applies Gaussian noise within patches (Lopes et al., 2019). Cutmix randomly cuts a portion from one image and pastes it onto another image, while assigning labels as a convex combination of the labels from the two images (Yun et al., 2019). Other studies employ a combination for partial detection (Kortylewski et al., 2020), or adopt a multi-task learning approach to predict the mask (Heo et al., 2022). Differing from these methods, our approach involves assigning one-hot labels to occlusion, treating it as a virtual class. When we occlude images with random noise, the labels are fused accordingly. This aids in generalizing better to common corruptions and occlusions compared to aforementioned methods. Section 3 provides further details on VF-training, and differences are depicted in Fig. 2.

3 Virtual Fusion

Let $\mathcal{X} = \{x_i\}_{i=1}^N$ denote the set of training samples and $\mathcal{Y} = \{y_i\}_{i=1}^N$ the corresponding labels, represented by one-hot vectors. Let $f : \mathcal{X} \rightarrow \mathcal{Y}$ be a classifier that maps each input $x_i \in \mathcal{X}$ to a label $y_i \in \mathcal{Y}$. Furthermore, consider $\mathcal{T}$ as a transformation that, when applied to x_i , introduces specific distortions to the input data. In the standard augmentation methods, the transformed data $\mathcal{T}(x_i)$ is incorporated into the training process and

associated with the originally assigned label y_i . In the VF-training, we think of transformations as virtual classes, and associate each transformation with a one-hot encoded virtual labels. This means that alongside the existing labels $\mathcal{Y}$, we introduce a set of virtual labels $\bar{\mathcal{Y}}$. Given an image x_i labeled as y_i , and a distortion type labeled as $\bar{y}_j$, the resulting corrupted image is represented as $\mathcal{T}_j(x_i) = \bar{x}_{i,j}$, and the labels are combined as shown in Equation 1, where $\alpha \in [0, 1]$ controls the level of noise added to the input data.

$$\bar{y}_{i,j} = \alpha y_i + (1 - \alpha) \bar{y}_j. \quad (1)$$

The loss function $\mathcal{L}_{VF}$, defined in Equation 2, shares similarities with Mixup (Zhang et al., 2017) and Cutmix (Yun et al., 2019), with the distinction that the second term represents the virtual labels:

$$\mathcal{L}_{VF} = \alpha \mathcal{L}(f(\bar{x}_{i,j}), y_i) + (1 - \alpha) \mathcal{L}(f(\bar{x}_{i,j}), \bar{y}_j) \quad (2)$$

where $\mathcal{L}$ represents the cross-entropy loss.

4 Experimental Setup

The following provides details on the dataset, training configuration, and evaluation metrics. Afterwards, we present the results and analysis.

4.1 configurations and metrics

Datasets. We employ the CIFAR-10 dataset (Krizhevsky et al., 2009), comprising 50,000 training images and 10,000 testing images sized at $32 \times 32 \times 3$. To measure model robustness against common data shifts, we utilize the CIFAR-10-C benchmark (Hendrycks and Dietterich, 2019), consisting of 15 corruptions such as noise, blur, weather, and digital distortions, each at severity levels $1 \leq s \leq 5$. This selection includes corruptions like Gaussian Noise, Shot Noise, Impulse Noise, Defocus Blur, Glass Blur, Motion Blur, Zoom Blur, Snow, Frost, Fog, Brightness, Contrast, Elastic Transform, Pixelate, and JPEG Compression. Following (Zhong et al., 2020), to assess robustness to occlusion, we generate an occluded dataset by randomly masking images within the CIFAR-10 test dataset. The masks cover up to 30% of the total image area, and are filled with either zero values, single uniform colors, or random noise.

Metrics. In Tables 1, 2, the Clean Error denotes the standard classification error on uncorrupted test data. As CIFAR-10-C contain 15 corruptions, first, for a given corruption c , we compute the error rate at severity s as $E_{c,s}$. The corruption error CE_c is the average error across all severities: $CE_c = \frac{1}{5} \sum_{s=1}^5 E_{c,s}$. Finally, the mean Corruption Error mCE is computed as the average of all 15 corruption errors: $mCE = \frac{1}{15} \sum_{c=1}^{15} CE_c$. This provides a single value for comparing robustness against common corruptions (Hendrycks and Dietterich, 2019).

Training Configuration and Hyperparameters: We conduct all experiments on a GeForce RTX-3080Ti GPU with CUDA Version 12.0, using PyTorch version 2.0.1. We employ a 40-2 Wide ResNet architecture (Zagoruyko and Komodakis, 2016) for training. The initial learning rate is 0.1, and it decays by a factor of 0.0001 using a cosine annealing schedule (Loshchilov and Hutter, 2016). Prior to any augmentations, we preprocess input images through random horizontal flipping and cropping. In all experiments, we train for 50 epochs, optimizing with stochastic gradient descent with a momentum of 0.9. For both training and evaluation, we set the batch size to 1024.

Baseline comparisons. In Tables 1, 2, "Normal" means the model is trained on clean data. To assess the effectiveness of VF in addressing occlusion, we evaluate its performance in comparison to various methods including Cutout (DeVries and Taylor, 2017), Random Erasing (Zhong et al., 2020), Patch Gaussian (Lopes et al., 2019), and Cutmix (Yun et al., 2019). In the case of Cutout and Random Erasing, we randomly select scale values from the range of (0.02, 0.3), and ratio from the range of (0.3, 3). For Patch Gaussian, we implement it using a patch size of 10 and a maximum scale of 0.3. For Cutmix, we utilize the implementation provided by the Torchvision (Transform v2, 2024) library and set the $\alpha = 1$. In VF experiments, we use a setting

Error	Normal	Std. Aug.	VF Aug.
Clean	9.04	9.05	8.98
Occlude	26.53	24.24	22.44
MCE	21.58	18.49	14.24
Gauss.	25.78	17.52	11.89
Shot	22.83	16.19	11.28
Impulse	35.21	22.08	14.10
Defocus	15.61	13.21	10.98
Glass	33.26	31.07	17.86
Motion	23.25	19.97	14.23
Zoom	20.23	15.05	10.31
Snow	19.42	19.02	15.76
Frost	18.62	18.18	13.08
Fog	21.53	20.78	20.82
Bright	10.92	10.63	9.39
Contrast	26.90	25.37	24.84
Elastic	16.49	16.30	13.34
Pixel	18.23	16.75	12.96
JPEG	15.53	15.28	12.84

Table 1: Comparison of error rates: Normal vs. Standard Augmentation vs. VF Training on Wide ResNet for CIFAR-10-C dataset while using Brightness, Gaussian Noise, and Zoom Blur as augmentations.

similar to Random Erasing but with the addition of an extra virtual class—denoted as VF_1 in Tables 2—to address occlusion. Similarly, in experiment VF_2 , we incorporate an additional four virtual classes to account for Brightness, Gaussian Noise, Zoom Blur, and occlusion. In all experiments except for VF_1 and VF_2 , we adjust the last classification layer of the network to output 10 class categories corresponding to the CIFAR-10 dataset. For VF, we add additional units depending on the number of augmentations used to accommodate the prediction of virtual classes. Since neither CIFAR-10 nor CIFAR-10-C class categories are associated with augmentations such as occlusion, thus during the testing phase, we ignore the outputs for these labels and only consider the 10 class categories of CIFAR-10 as the final output of the models.

4.2 Results

Comparisons of VF versus regular data augmentation. Similar to the findings of (Mintun et al., 2021), we observe that while adding several dissimilar augmentations has an impact on errors, the reduction in errors from including a few similar augmentations is much more significant. In other words, whether through normal augmentation or VF, we found that performance improves more when the augmentation used closely matches the scenarios encountered during testing. Table 1 shows improvement gains using three similar distortions—Brightness, Gaussian Noise, and Zoom Blur—for training and testing on the CIFAR-10-C benchmark (Hendrycks and Dietterich, 2019). The results of testing the same trained model on occlusion are presented in row "occlude" of the Table 1. While it demonstrates some improvement using augmentation for occlusion, direct training on occlusion using VF proves to be more effective for enhancing robustness. This is presented in column VF_1 in Table 2.

Comparisons of VF versus occlusion augmentation. Table 2 presents errors from various occlusion handling strategies. Fig. 2 summarizes percentage changes in error rates. For fair comparison between occlusion handling techniques, VF_1 uses occlusion alone as augmentation and fills the masked areas with random noise. In contrast, VF_2 employs four types of augmentation: Brightness, Gaussian Noise, Zoom Blur, and occlusion. VF_1 marginally improves Clean error but enhances occlusion and mCE errors by 49.26% and 8.22%, respectively. It falls behind Cutmix in Clean error rate improvement but outperforms it in occlusion and mCE error. Overall, VF_2 achieves the most substantial improvement, which we attribute to both VF label adjustment and similarities between data distributions during train and testing.

Error	Normal	Cutout	Rand. Erase	Patch Gauss.	CutMix	VF ₁	VF ₂
Clean	9.04	9.81	9.91	9.55	8.55	9.02	8.75
Occlude	26.53	25.47	14.79	18.64	15.71	13.46	13.30
MCE	21.59	24.68	21.40	19.47	20.65	19.81	13.81
Gauss.	25.78	35.65	19.06	15.61	28.54	19.64	11.26
Shot	22.83	30.44	16.98	14.46	25.20	17.79	10.69
Impulse	35.21	45.61	21.74	18.18	34.42	21.23	13.27
Defocus	15.61	18.40	21.49	17.10	16.36	17.12	10.49
Glass	33.26	34.08	27.65	26.62	31.34	30.37	18.21
Motion	23.25	25.70	29.51	23.56	20.89	24.42	14.05
Zoom	20.23	23.09	27.53	20.11	21.40	21.88	9.87
Snow	19.42	20.46	19.17	19.41	17.44	18.39	15.06
Frost	18.62	21.42	18.63	18.43	18.00	19.00	12.87
Fog	21.53	22.59	24.99	22.98	16.82	21.45	20.46
Bright	10.92	12.24	12.38	11.68	10.10	11.11	8.65
Contrast	26.90	31.89	32.63	32.20	20.09	27.47	24.14
Elastic	16.49	17.61	19.93	17.48	15.78	17.06	12.80
Pixel	18.23	15.06	15.14	18.32	18.22	16.10	12.94
JPEG	15.53	15.93	14.24	15.87	15.14	14.15	12.45

Table 2: Error rates of occlusion handling strategies: Assigning more informative labels with occluded images outperforms normal augmentations for under-occlusion and common corruption scenarios.

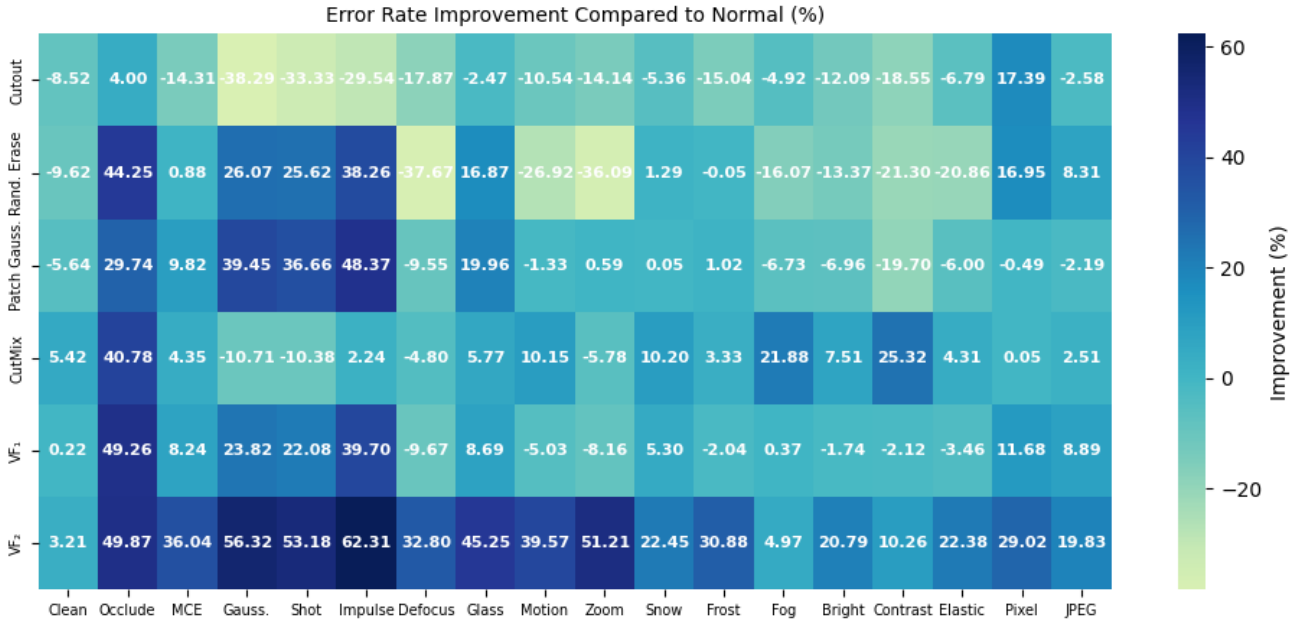


Figure 2: Error rate changes compared to standard training: While all methods enhance robustness in occlusion scenarios, techniques like Cutout, Random Erasing, and Patch Gaussian can increase the clean error rate. Except for VF₂, which exhibits similar distortion characteristics, errors under each corruption either decrease or increase compared to standard training. The mCE improvement in VF₂ highlights an emphasis on perceptual similarities.

5 Conclusion

Occlusion commonly arises in everyday visual tasks, and CNNs demonstrate vulnerability in such scenarios. Existing methods for addressing occlusions do not differentiate between the labels assigned to clean and occluded images despite some changes occurring to images in query. To enhance the informativeness of labels assigned to occluded images during training, instead of solely presenting the model with partially occluded image versions, we employed VF and assigned virtual labels for occlusion. This labeling strategy effectively communicates to the model the presence of occlusions in images, enabling it to differentiate between clean and occluded version. The findings suggest its efficacy compared to other occlusion handling methods and that perceptual similarities between training augmentations and test time scenarios could result in more robustness.

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